

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 1.3: Quadratic Equations — Teacher Reference (Pre-filled)

SUMMARY TABLE: MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 1.3: Quadratic Equations
Driving Question	DRIVING QUESTION: How can we use quadratic equations to find unknown dimensions and quantities that help our community solve real planning problems — like building the Harambee hall in Githurai?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Predict & Anchor Phenomenon: The Githurai Harambee Hall	What did we observe about the Githurai hall problem? Record the trial-and-error table results ($w = 8$ to 12 and areas $80, 99, 120, 143, 168 \text{ m}^2$), the finding that $w = 10$ gives exactly 120 m^2 , that the area grows by different amounts, and that the perimeter $4w + 4$ is linear.	What did we learn? Record that when an unknown is multiplied by itself we get a w^2 term and a quadratic equation; the general form $ax^2 + bx + c = 0$ with a not equal to 0 ; and that for the hall $w^2 + 2w - 120 = 0$, $a = 1$, $b = 2$, $c = -120$.	How can we explain why the committee got stuck? Write 2-3 sentences using the evidence: area = $w(w + 2)$ has $w \times w$, so it is not linear; the perimeter $4w + 4$ is linear but does not give area; guessing works only if we are lucky, so we need new methods for quadratic equations.
Lesson 2 Expanding and Simplifying Quadratic Expressions: From the Githurai Hall to the Kericho Tea Farm	We observed that multiplying two brackets such as $(w + 2)(w + 3)$ gives four products. In the area model of the Kericho plot $(x + 5)(x + 3)$, the pieces were x^2 , $5x$, $3x$ and 15 . The area $x^2 + 8x + 15 \text{ m}^2$ has an x^2 term, but the fencing perimeter $4x + 16 \text{ m}$ does not. At $x = 5$, the plot is 10 m by 8 m , with area 80 m^2 and perimeter 36 m .	We learned to expand a product of two binomials using the distributive law and FOIL (First, Outer, Inner, Last), then collect like terms. For example, $(w + 2)(w + 3) = w^2 + 5w + 6$ and $(2x - 1)(3x + 4) = 6x^2 + 5x - 4$. The answer always has the form $ax^2 + bx + c$, and signs must stay with the terms that follow them.	We can explain that the Githurai hall expression $w(w + 2)$ and the Kericho plot area $(x + 5)(x + 3)$ are quadratic because an unknown length is multiplied by another length that contains the unknown, creating an x^2 term. A perimeter adds lengths, so it stays linear. We now know how quadratic expressions are formed, but we still need a way to go backwards to find the brackets and later the width w of the hall.
Lesson 3 Factorisation of Quadratic Expressions: Undoing Expansion to Find the Hidden Hall Dimensions	What did we observe? Example: When we reversed FOIL, the two numbers in the brackets multiplied to give the constant and added to give the middle coefficient. $x^2 + 6x + 9$ became $(x + 3)^2$, $x^2 - 49$ became $(x - 7)(x + 7)$, and $2x^2 + 7x + 3$ became $(2x + 1)(x + 3)$.	What did we learn? Example: Factorising means writing an expression as a product of brackets, and it is the reverse of expanding. We can always check a factorisation by expanding it. Perfect squares and the difference of two squares are special patterns.	How does this help explain the Githurai hall? Example: The hall expression $w^2 + 2w - 120$ factorises as $(w + 12)(w - 10)$ using the factor pair 12 and -10 (product -120 , sum 2). The same method works for the Gikomba mandazi revenue $x^2 + 7x - 30 = (x + 10)(x - 3)$. We have the hidden

			brackets, but we still wonder how they give the width; that is the next step.
Lesson 4 Solving Quadratic Equations by Factorisation: Zero Product Property and the Githurai Harambee Hall	We observed that when the factorised hall equation $(w + 12)(w - 10) = 0$ is written, one of the brackets must be zero, which gives $w = 10$ or $w = -12$. We saw that $w = 10$ matches the 10 m by 12 m row (area 120 m ²) in the trial-and-error table, and that $w = -12$ would give negative dimensions. In the Kisumu fish pond, $w^2 + 3w - 40 = 0$ gave $w = 5$ or $w = -8$.	We learned the Zero Product Property: if $A \times B = 0$ then $A = 0$ or $B = 0$. To solve a quadratic equation by factorisation we write it in the form $ax^2 + bx + c = 0$, factorise, set each factor equal to zero, solve for the two solutions, and check each one in the context.	A quadratic equation can have two solutions, but context tells us which one to use. For the Githurai Harambee hall, $w = 10$ and $w = -12$ both satisfy the equation, but a width cannot be negative, so we reject -12 and conclude the hall has width 10 m and length 12 m (area $10 \times 12 = 120$ m ²). In the same way, the Kisumu fish pond has width 5 m and length 8 m ($5 \times 8 = 40$ m ²).
Lesson 5 Solving by Completing the Square: The Nakuru School Garden	We observed that $x^2 + 6x - 48 = 0$ (the Nakuru school garden, area 48 m ² , length 6 m more than the width) has no integer factor pair with product -48 and sum 6, so it does not factorise. We also observed with paper pieces and a diagram that an x by x square and two 3 by x rectangles need a 3 by 3 corner (area 9) to form a square of side $(x + 3)$.	We learned that $x^2 + bx = k$ can be rewritten as $(x + b/2)^2 = k + (b/2)^2$ and solved by taking square roots, giving two solutions. For the garden, $(x + 3)^2 = 57$, so $x = -3 \pm \sqrt{57}$, which is 4.55 or -10.55. Context tells us to reject the negative width.	The garden is 4.55 m wide and 10.55 m long, because a width cannot be negative and $4.55 \times 10.55 = 48$ m ² (approximately). Completing the square works when factorisation fails, and it also gives the hall answer: $(w + 1)^2 = 121$, so $w = 10$ or -12, and the Githurai hall is width 10 m, length 12 m.
Lesson 6 The Quadratic Formula and the Mombasa Stone Problem	What did we observe? Record the values of a , b , c and D for the Mombasa stone ($-5t^2 + 20t + 25 = 0$), the Githurai hall ($w^2 + 2w - 120 = 0$), Mama Wanjiru's mandazi ($p^2 - 30p + 175 = 0$), $x^2 - 6x + 9 = 0$ and $x^2 + 2x + 5 = 0$, and the number of real solutions each gave.	What did we learn? Write the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ and explain how it comes from completing the square on $ax^2 + bx + c = 0$. State what $D > 0$, $D = 0$ and $D < 0$ mean for the number of real solutions.	How does this help explain the phenomenon? Explain how the formula gives the Githurai hall as width 10 m and length 12 m ($D = 484$, $w = 10$ or -12, reject -12), and how context decides which solution to use in the stone problem ($t = 5$ s) and Mama Wanjiru's pricing (7.93 KES chosen over 22.07 KES).
Lesson 7 Application, Model Building and Consolidation: Closing the Githurai Harambee Hall Story	We observed that the Githurai hall equation $w(w + 2) = 120$, or $w^2 + 2w - 120 = 0$, gives the same solutions, $w = 10$ or $w = -12$, by factorisation, completing the square and the quadratic formula ($D = 484$). New problems gave Kitale land $w = 6$ m (length 13 m), Harambee tickets $p = 10$ or 20 KES, and a verandah of $x = 1$ m.	We learned that a quadratic equation has the unknown multiplied by itself (highest power 2), unlike a linear equation such as the perimeter $4w + 4$, and can have two solutions. We learned how to choose the most efficient method and to check each solution in the context.	We explain that the hall must be width 10 m and length 12 m (area 120 m ²) because $w = -12$ is impossible for a width, so context decides which solution is used. Quadratic thinking lets the Harambee committee and our community find unknown dimensions and quantities for real planning problems.

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